

Creative Computing with AI

Course: AI 201 | **Weight:** 85% (across 6 milestones) | **Due:** Milestones due at checkpoint weeks; final demo Week 10

AI use: Open from Day 1. Document everything. ESF escalates at Milestone 2.

Purpose

Code is a creative medium. AI is a production partner. This course asks whether you can direct both with intention, and whether you understand what you built.

You will spend the full quarter on one creative direction: exploring it, researching it, mapping its system, building it, breaking it, rebuilding it, and presenting it. The direction is yours. The tools are yours. The creative decisions are yours. The course teaches the thinking, the workflow, and the discipline of ownership.

By Week 10 you will have a working system you can explain, a process book that proves you understand it, and a live demo that shows what it does. The question is not whether AI can build it for you. The question is whether you can direct AI to build something that is genuinely yours.

Your Quarter Question

Write one question you want to answer this quarter:

"What happens when I use [creative computing concept] to [do something] in [my discipline]?"

Examples:

- "What happens when I use generative systems to create layout compositions for editorial design?"
- "What happens when I use code to generate procedural environments from AI-written scripts for game worlds?"
- "What happens when I use interaction design to let dancers control live visuals with their bodies?"
- "What happens when I use algorithmic patterns to generate textile designs for fashion?"

This question carries through every milestone. It can evolve. The evolution is documented. By Milestone 6, you answer it. See [quarter-question](#) for guidance on writing and testing your

question.

Milestones

Six checkpoints across the quarter. Each milestone follows the Agile workflow: Declare (state what you intend), Make (build it), Show (get feedback), Document (capture what happened and why).

Milestone 1: Create + Explore (Weeks 1 to 2)

Build something. Use AI, use code, vibe code it: get a working prototype on screen by the end of Week 1. Do not plan it. Make it.

Then take it apart. In Week 2, you deconstruct what you built: what does the code actually do? Where did AI make decisions you did not notice? What would you change if you understood every piece? This is the gap between having a thing and owning a thing. The rest of the quarter closes that gap.

Checkpoint deliverables:

- Working prototype (whatever you built)
- Quarter question (written)
- Process blog entries documenting every AI interaction
- CLAUDE.md and README created in your repo

What this milestone tests: Can you make something? Can you identify what you do not yet understand about it?

Milestone 2: Research to Verify (Week 3)

Now that you have built something, you know what questions to ask. Research with purpose.

Verify your direction: find at least 3 practitioners or projects working in your area. Evaluate your tools: is the environment you used in M1 the right one, or did you default to the first thing that worked? Write your Position Statement: you now have a direction worth defending.

Checkpoint deliverables:

- Research documentation (3+ practitioners, tool evaluation)
- Position Statement (3 elements: stance, what matters most, what I will not compromise on)

- Design Intent with system sketch (input, process, output) — see [design-intent](#)
- Process blog continues

What this milestone tests: Can you ground your direction in research? Can you articulate what you are building and why?

ESF enters here. The Position Statement is your reference point for every decision from this point forward. AI use becomes directed: you are no longer exploring freely. You are building toward a stated intent.

Milestone 3: System Map + Leverage Points (Weeks 4 to 5)

Map the system around your project. What are the components? How do they connect? Where does code actually matter?

Identify the **leverage points**: the places in your system where a small change produces a large effect. These are the lines of code, the model parameters, the interaction patterns that make or break the experience. You will document these in your process book with code snippets explaining what each one does and why it matters.

Build the first version of your system map. This is not decorative. It is an architecture document showing perceive, decide, act: what the system senses, how it processes, what it produces, and where you are in the loop.

Checkpoint deliverables:

- System map (perceive, decide, act pipeline)
- Leverage points identified with code snippets and explanations
- First Playable: the system works end-to-end, even if rough
- Process blog continues

What this milestone tests: Do you understand your system well enough to map it? Can you identify and explain the code that matters most?

Milestone 4: Prototype v1, Break, Iterate to v2 (Weeks 6 to 7)

Build prototype v1 from your system map. Then break it.

"Breaking" means deliberately testing the limits: what happens when the input changes? What fails when you remove a component? What does the system do that you did not intend?

Breaking the prototype is how you discover what you do not yet understand. It is also how you find new creative possibilities you would not have planned.

Iterate to v2 based on what breaking revealed. Test v2 with someone who is not you. Document their experience.

Checkpoint deliverables:

- Prototype v1 (working)
- Documentation of what you broke and what you discovered
- Prototype v2 (iterated based on findings)
- User testing documentation — see [user-testing](#)
- Records of Resistance begin (minimum 1 by end of M4)
- Process blog continues

What this milestone tests: Can you build, break, and rebuild? Can you learn from failure and from outside feedback?

Milestone 5: Prototype v3 + Integrated Feedback (Weeks 8 to 9)

Integrate everything: user testing feedback, findings from breaking, new technical discoveries, and your own evolving understanding.

Prototype v3 is not a polish pass. It is the version where every decision is intentional and you can explain why. The leverage point code snippets in your process book should now show the full evolution: what the code did in v1, how it changed in v2, and what it does in v3.

Update your system map to reflect the final architecture.

Checkpoint deliverables:

- Prototype v3 (working, tested, intentional)
- Updated system map (final architecture)
- Leverage point evolution documented (v1 to v3 code snippets with explanations)
- Records of Resistance (minimum 3 cumulative by end of M5)
- Process blog continues

What this milestone tests: Can you integrate feedback and iteration into a system you fully own? Does the process book show the arc from M1 to here?

Milestone 6: Process Book + Final Demo (Week 10)

Complete your process book. Present your work.

The process book is the evidence of your quarter. It contains your quarter question and how it evolved, your research, your system maps, your leverage point code snippets with explanations, your iteration arc from M1 prototype to v3, your Records of Resistance, and your reflection on what you learned.

The final demo is live. Show the system working. Walk through the pipeline. Explain the leverage points. Answer your quarter question.

Checkpoint deliverables:

- Process book (complete)
- Final playable demo (live, presented to the class)
- Artist/designer statement — see [artist-designer-statement](#)
- Five Questions (applied honestly)
- Disclosure Statement
- CLAUDE.md final update + repo close-out

What this milestone tests: Can you present a system you built, explain how it works, and demonstrate that the creative decisions are yours?

The Process Book

The process book is the central artifact of this course. It runs from Week 1 through Week 10 and documents everything: AI interactions, creative decisions, iteration, failure, discovery. What distinguishes it from a blog is code snippets at leverage points (explaining what the code does and why it matters), a visible iteration arc from M1 through v3, and entries that record decisions rather than just actions.

See [process-blog](#) for the full guide.

ESF Escalation

Milestone	ESF Status	What Is Required
M1 (Weeks 1 to 2)	Open	Document all AI interactions in process blog. No formal ESF.

Milestone	ESF Status	What Is Required
M2 (Week 3)	Position Statement	Write Position Statement before directed building begins. AI use becomes intentional.
M3 to M4 (Weeks 4 to 7)	Active	Records of Resistance begin. Minimum 1 by end of M4.
M5 (Weeks 8 to 9)	Full	Minimum 3 Records of Resistance cumulative.
M6 (Week 10)	Reflective	Five Questions. Disclosure Statement. Process book shows the full ESF arc.

Assessment

85% of your final grade, distributed across milestones.

Milestone	Weight	What Is Evaluated
M1: Create + Explore	10%	Prototype exists. Process blog started. Quarter question written.
M2: Research to Verify	5%	Research grounded. Position Statement specific. Design Intent clear.
M3: System Map + Leverage Points	10%	System map accurate. Leverage points identified with code explanations. First Playable works.
M4: Prototype v1, Break, v2	25%	Breaking produced discoveries. v2 shows iteration. User testing documented.
M5: v3 + Integrated Feedback	15%	v3 integrates all feedback. Leverage point evolution documented. System map updated.
M6: Process Book + Final Demo	20%	Process book complete. Demo works. Statement placed. Presentation clear.

Participation (15%) is assessed separately: studio check-ins, group check-ins, peer feedback, crit engagement.

Four evaluation dimensions across all milestones:

Dimension	What It Means
Creative vision	Strength of the concept. Intentional choices. Connection to your discipline. Does the project answer your quarter question?

Dimension	What It Means
Technical ownership	Does the system work? Do you understand why? Can you explain the leverage points? Can you modify the system without starting over?
Process and documentation	Quality of the process book: AI interactions documented, iteration visible, code snippets at leverage points, reflection specific.
Critical thinking	Position Statement defended. Records of Resistance showing editorial judgment. Five Questions applied honestly. Awareness of your own creative process.

Repo Structure

One project, one repo. The repo is your portfolio artifact.

```

  toolkit-repo/
  |
  |   ← you create these
  |   └─ README.md           # For humans: who you are, what you built
  |   └─ CLAUDE.md           # For AI tools: your patterns, constraints,
  |   direction
  |   |
  |   |   ← ESF Companion installs these
  |   |   └─ .claude/
  |   |   └─ prompts/
  |   |   └─ templates/
  |   |
  |   |   ← your project work
  |   |   └─ process-blog/    # The central artifact
  |   |       └─ week-01.md
  |   |       └─ week-02.md
  |   |       └─ ...
  |   |       └─ leverage-points/ # Code snippets with explanations
  |   └─ project/
  |       └─ src/
  |       └─ assets/
  |   └─ research/           # Practitioner research, tool evaluations
  |   └─ system-maps/        # System map iterations (M3, M5 update)
  |   └─ esf/                # Position Statement, Records of Resistance,
  |   Five Questions

```

Resources

Supplement guides for each milestone are linked from the session docs. Full reference on the course microsite.

Submission

Milestone checkpoints are graded at the end of their designated weeks. Your repo is your submission. Keep it clean, navigable, and up to date.

Final submission (M6): Process book complete, demo presented in class, all ESF documents filed, CLAUDE.md updated, repo closed out.

Naming convention: AI201_[Last]_[First]_[Artifact].ext